Chapter 1.2 Distributions

Rick Durrett Probability: Theory and Examples

Hu Jingwu

Built from the local Chapter 1 LLMwiki flash cards.

Roadmap for 1.2

Learning goal

Move between laws, CDFs, densities/mass functions, atoms, and transformations.

1. Distribution functions and probability laws.
2. Atoms as jumps and countability of discontinuities.
3. Normal tail estimates.
4. Probability integral transform.
5. Density transformations: monotone maps, lognormal, and squares.

Distribution Function Properties

Theorem

If $F(x) = \mathbb{P}(X \leq x)$, then:

$$F \text{ is nondecreasing}, \qquad \lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to \infty} F(x) = 1,$$

and F is right-continuous.

Pitfall

A CDF need not be left-continuous. Its jumps are atoms.

LLMwiki: `Distribution function properties; ID: durrett_ch1_distribution_function`

How to Prove CDF Properties

Proof idea

Translate analytic properties of F into event inclusions and continuity of probability.

LLMwiki: `Distribution function properties; ID: durrett_ch1_distribution_function`

Every Valid CDF Determines a Law

Theorem

Any function $F : \mathbb{R} \to [0, 1]$ that is nondecreasing, right-continuous, and satisfies

$$F(-\infty) = 0, \qquad F(\infty) = 1$$

is the CDF of a unique probability measure on $\mathcal{B}(\mathbb{R})$.

- Define $\mu((a, b]) = F(b) - F(a)$.
- Extend from half-open intervals to the Borel sigma-field.

LLMwiki: `Every valid CDF determines a law; ID: durrett_ch1_cdf_to_law`

Atoms Are Jumps

Fact

For a real random variable with CDF F,

$$\mathbb{P}(X = x) = F(x) - F(x-), \qquad F(x-) = \lim_{y \uparrow x} F(y).$$

LLMwiki: `Atoms are jumps of the CDF; ID: durrett_ch1_atoms_jumps`

Standard Normal Tail Estimate

Inequality

If χ is standard normal and $x > 0$, then Durrett's normal-tail estimate gives

$$(x^{-1} - x^{-3})e^{-x^2/2} \leq \int_x^\infty e^{-y^2/2}\,dy \leq x^{-1}e^{-x^2/2}.$$

Hence

$$\mathbb{P}(\chi \geq x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-y^2/2}\,dy.$$

LLMwiki: `Standard normal tail estimate; ID: durrett_ch1_normal_tail_bound`

Probability Integral Transform

Fact

If X has continuous CDF F, then

$$F(X) \sim \mathrm{Unif}(0,1).$$

- Proof route: compute $\mathbb{P}(F(X) \leq y)$ through a quantile.
- Flat intervals require care, but continuity makes their probability contribution zero.

LLMwiki: `Probability integral transform; ID: durrett_ch1_probability_integral_transform`

Monotone Density Change of Variables

Theorem

Suppose X has density f on $[\alpha, \beta]$ and g is strictly increasing and differentiable. If $Y = g(X)$, then

$$f_Y(y) = \frac{f(g^{-1}(y))}{g'(g^{-1}(y))}\mathbf{1}_{\{g(\alpha)<y<g(\beta)\}}.$$

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.1

Let X and Y be random variables on $(\Omega, \mathcal{F}, P)$ and let $A \in \mathcal{F}$. Define $Z = X$ on A and $Z = Y$ on A^c. Show that Z is a random variable.

LLMwiki: `Piecewise random variable measurability; ID: exercise_method_piecewise_random_variable_measurability`

Exercise 1.2.1: Proof Skeleton

LLMwiki: `Piecewise random variable measurability; ID: exercise_method_piecewise_random_variable_measurability`

Exercise 1.2.2

Let χ be standard normal. Use Durrett's normal-tail estimate to find upper and lower bounds for $\mathbb{P}(\chi \geq 4)$.

LLMwiki: `Standard normal tail estimate; ID: durrett_ch1_normal_tail_bound`

Exercise 1.2.2: Proof Skeleton

LLMwiki: `Standard normal tail estimate; ID: durrett_ch1_normal_tail_bound`

Exercise 1.2.3

Show that a distribution function has at most countably many discontinuities.

LLMwiki: `Countability of CDF jumps; ID: exercise_method_countable_cdf_discontinuities`

Exercise 1.2.3: Proof Skeleton

LLMwiki: `Countability of CDF jumps; ID: exercise_method_countable_cdf_discontinuities`

Exercise 1.2.4

Suppose $F(x) = \mathbb{P}(X \le x)$ is continuous. Show that $Y = F(X)$ is uniform on $(0, 1)$, i.e.

$$\mathbb{P}(Y \le y) = y, \qquad 0 \le y \le 1.$$

LLMwiki: `Probability integral transform; ID: durrett_ch1_probability_integral_transform`

Exercise 1.2.4: Proof Skeleton

LLMwiki: `Probability integral transform; ID: durrett_ch1_probability_integral_transform`

Exercise 1.2.5

Suppose X has continuous density f and is supported on $[\alpha, \beta]$. Let g be strictly increasing and differentiable on (α, β). Find the density of $g(X)$.

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.5: Proof Skeleton

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.6

Suppose X is normally distributed. Use the previous exercise to find the density of $\exp(X)$, the lognormal distribution.

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.6: Proof Skeleton

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.7

(i) If X has density f, compute the distribution function and density of X^2.

(ii) Specialize to X standard normal to get the chi-square density with one degree of freedom.

LLMwiki: `Density of a square transform; ID: exercise_method_square_transform_density`

Exercise 1.2.7: Proof Skeleton

LLMwiki: `Density of a square transform; ID: exercise_method_square_transform_density`

Checklist: How to Attack 1.2 Problems

1. Need to verify a CDF? Check monotonicity, right-continuity, and endpoint limits.
2. Need an atom? Compute $F(x) - F(x-)$.
3. Need countability of jumps? Group jumps by size and bounded interval.
4. Need a transformed density? Compute the transformed CDF first, then differentiate.
5. Need a normal tail? Keep the exponential term and constants separate.

LLMwiki: `Chapter 1.2 flash cards and exercise method cards`

Mini Practice

1. If F is continuous and strictly increasing, show $\mathbb{P}(F(X) \leq y) = y$ directly using F^{-1}.
2. If X has density f, find the density of $aX + b$ for $a < 0$.
3. Show that a CDF can have only finitely many jumps larger than 0.01.
4. If $X \sim N(0, 1)$, use the tail bound to estimate $\mathbb{P}(|X| \geq x)$.